

Series B Series B

Series B • Hands-on

Summary and reference cards

What is worth keeping once all nineteen episodes are done.

Screenshot one card, or print the whole set and pin it next to your screen.

B1-B19 condensed | 36 cards | Every number measured, not invented

The Series B map: nineteen episodes on one card

Four stages, three handoffs: upstream hands over the count matrix, core the annotated object, applications the DE results — each handoff is an RDS file

Upstream B1-B4



FASTQ → count matrix

B1 Setup plus your first UMAP | B2 mkref and FASTQ anatomy |
B3 count and reading web_summary | B4 aggr / multi / Feature
Barcode

Core B5-B10



Matrix → a named cell map

B5 QC | B6 normalization and HVGs | B7 PCA | B8 clustering |
B9 UMAP | B10 annotation — PBMC 3k throughout

Applied B11-B12



Multi-sample → trustworthy differences

B11 integration and batch diagnosis | B12 pseudobulk + DESeq2
— the key statistics lesson; ifnb carries this branch

Advanced B13-B19



The "so what" after the list

B13 enrichment | B14 communication | B15 trajectories | B16
inferCNV | B17 deconvolution | B19 strategy | B18 finale

01

Upstream • Cell Ranger

B1–B4: half of your result is decided before count even starts

B1 • Environment setup and your first UMAP

In one line: install today, get your first UMAP today — the environment is part of your lab equipment



Verify versions

R 4.3+ and Seurat v5 — see the 5.x with your own eyes via `packageVersion`; lock everything with `renv` so results reproduce



Keep paths relative

RStudio Project plus relative paths; no `setwd` and no absolute path appears anywhere in this series



The whole run in 15 lines

Read10X to DimPlot: 2,638 cells after filtering, 9 clusters. Matching numbers means reproduced — a mirrored UMAP is the same plot



Trap in one line

Installing the latest blindly (v4 tutorial code breaks on v5 Layers); a hard-coded `setwd` dies on line one of any other machine

B2 • References and FASTQ: mkref and mkfastq

In one line: FASTQ answers who, what and which sample; the reference decides which genes can be counted at all



Three files

R1 = barcode + UMI (who), R2 = cDNA (what — the only one that gets aligned), I1 = sample index (which sample)



A reference is two things

FASTA governs alignment, GTF governs counting — versions must match. Same FASTQ, two references: 2,300 genes apart (20,300 vs 22,600)



The transgene ritual

Add GFP to both FASTA and GTF; after building, one grep must find the gene name in the GTF before count is allowed to run



Trap in one line

Loud traps (mixed versions, aligning R1 as cDNA) blow up on the spot; the silent one costs most — GFP missing from the GTF counts as an exact zero, forever, with no warning

B3 • cellranger count and reading web_summary

In one line: count produces a report, not an answer — and you are the one who signs off on it



Four gates

Warnings, cell count (against the loading record), knee-plot shape (shelf, cliff, plateau), and the mapping breakdown. All four, or it stays upstream



Healthy ranges

Valid Barcodes > 97%; Mapped to Genome 80–90%. Cells: exonic ~55%, intronic ~28% — intronic flipping past 50% means cytoplasm loss (snRNA-seq excepted)



Saturation is economics

At 32% saturation, doubling reads adds 43% more UMIs; doubling again adds 19%. Ask whether median genes answers your question, not whether the percentage passes



Trap in one line

Reading web_summary numbers as isolated scores — a number without a reference point is not interpretation, it is mood

B4 • Multi-sample and multimodal: aggr, multi, Feature Barcode

In one line: count runs one sample, multi runs one experiment — and merging defaults to downstream



Three kinds of multi

Multiple lanes: count handles them. Multiple samples: run count per sample, merge in R by default. CMO/VDJ go through multi; ADT/HTO also work as count + Feature Barcode



What aggr costs

Depth normalization downsamples your best sample to match the worst — usually turn it off and let downstream statistics handle depth



Three hashing calls

Demultiplexing yields singlet / doublet / negative; do not discard negatives wholesale — some populations simply do not stain with HTOs



Trap in one line

Swap the GEX and antibody FASTQ pairing and both mapping rates crash together — when both sides look bad at once, check the libraries CSV first

02

Core and applied • Seurat

B5–B10 build the cell map; B11–B12 make multi-sample differences trustworthy

B5 • Loading data and QC

In one line: look at the distributions first, then set thresholds, then record the reasons — answers change with data, the procedure does not



The measured numbers

13,714 genes by 2,700 cells; 2,638 remain after filtering — 62 cells, about 2.3%, removed. Log what every step discards



Two ways to draw the line

Fixed cutoffs (here nFeature 200–2,500, mt < 5% — do not port them to other tissues) versus median $\pm 3 \times \text{MAD}$; cross-check both, and save pbmc_raw.rds before filtering



Handling doublets

Every extra thousand cells adds ~0.8% doublets; DoubletFinder marks without removing — labels go into metadata and the reckoning happens after clustering



Trap in one line

A blanket 5% mt cut (goodbye cardiomyocytes); filtering before looking (order reversed); one-pass QC (debris clusters only surface after clustering)

B6 • Normalization, HVG, scaling

In one line: three steps for three problems — normalization for depth, HVG for noise, scaling for magnitude; the route matters less than having a reason and finishing it

→ Two routes

LogNormalize (CP10K + log1p) → vst picks 2,000 HVGs → ScaleData (z-score + clip); or SCTransform, all three steps in one line



The measured face-off

HVG overlap between routes: 1,245 of 2,000 (just over 60%); the major-population structure and canonical-marker conclusions agree completely



The lifesaver check

After PCA, correlate PC1 with nCount — one line, three seconds, and it blocks the most expensive disaster there is



Trap in one line

Skip normalization and PC1 becomes the sequencing-depth axis; log again after SCT and the double transform flattens every marker into grey

B7 • PCA and choosing dimensions

In one line: three methods frame an interval, settle on the generous side, write the reason down — then stop worrying



Three ways to pick nPC

The elbow sits at 7–12, the quantified cutoff points to PC13, the stability scan confirms the plateau — this series settles on 1:10



Forgiving, but asymmetric

nPC 10 and 30 are nearly the same picture; at nPC = 5 two T subsets fuse and vanish — subset signal often lives in weak mid-range PCs like PC7



Copy-ready methods text

"Top ten PCs were used. The elbow falls at PCs 7–12; adjacent variance differences drop below 0.1% at PC 13."



Trap in one line

Too few PCs is the only fatal mistake — subsets vanish silently, no error raised; if PC1 is depth or ribosome (measured $r = 0.98$), go back and fix normalization

B8 • Clustering: the resolution scan and clustree

In one line: clustering hands you hypotheses — clustree finds the stable band, markers decide life and death, and the reason gets written down



The measured scan

Resolution 0.2→1.4 yields 6, 8, 10, 11, 13, 15, 16 clusters — same data, nearly a threefold range. Cluster count is asked for, not revealed



How the verdict works

Stable band 0.4–0.8; every cut at 0.6 (ten clusters) has marker support, while 0.8's extra cut tops out at RPL/RPS genes with pct 99% on both sides — a state gradient, not a type. Verdict: 0.6



The pre-verdict autopsy

Check per-cluster QC medians and doublet fractions; platelets are naturally low-nFeature — no single metric convicts, always read it with markers



Trap in one line

Tuning resolution until the desired count appears (circular); across runs trust cross-tables, never label numbers; debris has a marker list too — forming a cluster is not biology

B9 • UMAP / t-SNE: parameters, randomness, reproducibility

In one line: analysis lives in PC space, pictures live on the UMAP; nail the seed, write down the parameters



Three levels of seed

`set.seed(1234)`, RunUMAP's `seed.use`, and single-threading plus recorded versions — handle all three and your figure reproduces



Two knobs

`n.neighbors` 5/30/100 trades local for global; `min.dist` 0.01/0.3/0.8 sets within-cluster tightness — squeeze `min.dist` and any dataset looks well separated



Distance flips, measured

Change the seed: cluster 4-to-1 distance goes 7.1→14.1 while 4-to-5 goes 16.7→7.6; rank correlation across all 15 pairwise distances is $\rho = -0.04$ — a coin flip



Trap in one line

Only one thing on a UMAP is trustworthy: who belongs to each cluster. Distance, area, shape and whitespace are stage scenery — coordinates never enter statistics

B10 • Annotation: FindAllMarkers, SingleR, Azimuth

In one line: manual builds the hypothesis, automation verifies it; when they argue, the evidence hierarchy rules — canonical markers > reference similarity > enrichment guesses



Reading order for the table

avg_log2FC first for effect size, then pct.1 versus pct.2 for exclusivity, p_val_adj last as gatekeeper — never sort by p value



What a good marker looks like

CD79A: pct 0.94 vs 0.04; NKG7: 0.99 vs 0.13 with log2FC 4.05. Counterexample B2M: 1.00 vs 0.99 — everyone expresses it, and no p value makes it a marker



Three-way cross-validation

Manual panel (CD3E, MS4A1, CD14...) + SingleR + Azimuth; adjudicate each disagreement, final labels into metadata, original numbers kept



Trap in one line

Smallest p is not best marker; auto labels are not conclusions (missing types get force-fitted the nearest name); lineage markers cannot split subclusters — CD3E tells four T subsets nothing

B11 • Integration and batch-effect diagnosis

In one line: look at the unintegrated baseline first, then integrate — and accept the result only if the canary survives



Three methods, one interface

IntegrateLayers swaps methods with one argument: RPCA conservative, Harmony fast (theta defaults to 2 — turn it down if overmixed), CCA strong but most prone to overcorrection; under ~100 cells per sample, lower k.weight



Three acceptance checks

split.by — same cluster, same position; per-cluster condition ratios (healthy is roughly 46:54 to 55:45); known markers lighting the same spot in both conditions



The canary test

A condition-specific population you already know exists must remain its own cluster after integration. Dead canary = overcorrected, redo it. A perfectly mixed map is not full marks — it is an alarm



Trap in one line

No unintegrated baseline, no way to judge what integration changed; integrating everything unconditionally tears rare populations apart; corrected values in DE — the rule: geometry uses integration, statistics returns to RNA

B12 • Differential expression: the FindMarkers trap and pseudobulk + DESeq2

In one line: sum the cells back into samples, and hand the statistics back to the samples



The naive approach, measured

FindMarkers on stim vs ctrl: 3,187 significant genes, p values hitting $1e-320$ — from an experiment with 16 people. Cells share donors and batches; the effective n is 8 versus 8



The right way

AggregateExpression (RNA counts summed per sample by type) → DESeq2 → lfcShrink: 1,024 up plus 906 down, about 1,930. On simulated data it is starker still: naive 303 versus pseudobulk 106



Three gatekeepers

RNA counts (right data layer), honest n (real samples), effect size and expression (plot per-sample dots for every shortlisted gene — a log2FC of 3.9 propped up by baseMean 2.1 is no headline)



Trap in one line

DE on integrated values (DESeq2 throws a negative-values error — that error is a gift); forcing n = 2 vs 2 and claiming conclusions; reading volcanoes by p alone, ignoring FC and expression

03

Advanced downstream • B-IV (B13–B19)

B13 list to conclusion, B14 who talks to whom, B15 the timeline of states, B16 who is malignant, B17 feeding bulk cohorts, B19 a strategy per tumor, B18 the map unfolded — tools always answer; you judge whether the answer exists

B13 • After DE: enrichment and gene-set analysis

In one line: GSEA asks direction, ORA asks the list, module scores ask location — the three layers must agree



Two routes

GSEA takes the ranking of all genes (rank by the statistic, no significance cutoff); ORA takes a list plus a background. A short list is the cue to switch to GSEA — turning B12's 1,930 genes into one biological sentence



Known-answer validation

fgsea + Hallmark on ifnb CD14 monocytes: top hit INTERFERON_ALPHA_RESPONSE at NES 2.88, runner-up IFN- γ at 2.79 — the answer is written into the experiment; if it is not on top, audit the pipeline, not the data



Report themes, not entries

GO parent-child double counting turns "top 20 entries" into about three findings; simplify away redundancy and converge by theme (this dataset has three: interferon antiviral, antigen presentation, chemokines) — NES direction must agree within a theme



Trap in one line

The background trap costs most: set the universe to the whole genome instead of detected genes and all seven test entries go "significant" — the four extras (ribosome, translation...) are all fake; asking what your own marker list enriches for is circular reasoning

B14 • Cell-cell communication: ligand-receptor and CellChat

In one line: CellChat turns expression into communication hypotheses worth testing — strong at generating them, not enough as evidence



The chain of steps

createCellChat → identifyOverExpressed → computeCommunProb (triMean needs ~25% of cells expressing) → aggregateNet; build CTRL and STIM objects separately with identical parameters, compare only after mergeCellChat



Granularity: 8 vs 14 flips it

Same data: at 8 clusters the story is "monocytes send CXCL10 to NK"; at 14, activated T stands alone (5% of T cells) and the strongest axis becomes monocytes-to-activated-T — rerun key conclusions at a second granularity



Report p and probability as a pair

The permutation p only says "above random" — at nboot 100 its resolution is 1/100, a whole row bottoms out with no ranking power; strength lives in the communication probability. min.cells 10 is the floor — tiny groups roll dice



Trap in one line

None of the three assumptions is verified — RNA ≠ protein ≠ secretion ≠ actual contact; communication is inference, not observation, and the verb in your conclusion is always "suggests"

B15 • Trajectory analysis and pseudotime

In one line: a trajectory answers how cells got there; whether they are going anywhere is yours to answer first — the tool always draws a line



Three qualifying questions

Is the process continuous? Were intermediates sampled? Is the root biologically grounded? Force discrete PBMC types through and you still get a pretty "differentiation path" — branch points at type boundaries are artifacts



Root reversal, measured

Rerun Slingshot with a different root: rank correlation between the two pseudotimes is -0.998 , the whole ordering flips and the maturation marker's smoother turns from rising to falling — the direction is wrong and nothing ever warns you



Compute in PCA, look at UMAP

Rank correlation with true time: 0.99 in PCA space, 0.98 on UMAP coordinates — the small gap is this simulation's luck; UMAP also randomly tears the trunk into an island, local order gets shuffled, and pseudotime quality is left to the seed



Trap in one line

Pseudotime is arc length along the curve — an ordering, not a clock; computing it on UMAP coordinates and calling it quantitative forgets everything B9 said about untrustworthy distances

B16 • Tumor scenes: inferCNV and calling malignant cells

In one line: malignancy is read from the genome's shadow — sort, window, compare to baseline, then triangulate



The chain of reasoning

Sort genes by genomic position → sliding-window averaging cancels single-gene noise → chromosome-scale shifts relative to the reference; the chr1p/19q deletion in the built-in oligodendroglioma example is the textbook signal



Wrong reference zeroes it out

Measured with a suspect-malignant group as baseline: its own deletions zero out into blank rows, while the truly normal cells sprout fake "gains" at 1p/19q — the direction fully reverses, and the wrong heatmap looks flawless



Three sources of fake signal

A misplaced baseline; co-regulated HLA/IG clusters smoothed into fake amplifications (simulated pure-diploid cells still grow a red hill on chr6); near-diploid tumors with clean CNV — the inference is one-way: events all but prove malignant, absence proves nothing



Trap in one line

CNV is one piece of evidence, not a verdict — call cells by triangulating CNV, clusters and markers; put only trusted immune and stromal cells in the reference, and the first glance at any heatmap checks whether the reference rows are clean

B17 • Deconvolution: unmixing bulk with a single-cell reference

In one line: grade yourself on fake bulk with known proportions before unmixing the real bulk you have no answer key for



One equation, two directions

bulk $\approx$ signature $\times$ proportions: B12 sums back, B17 unmixes back. From pbmc3k annotations, a 138-gene $\times$ 6-type signature (CD4-CD8 column correlation 0.8, on record), 20 fake bulks at known proportions, NNLS against the key



The overall score lies

Overall $r = 0.992$, RMSE 0.02 — propped up by big types like CD4 and monocytes; split per type and rare populations under 5% carry markedly higher error. Abundant types land, rare types drift



Missing type = next-of-kin steps in

Remove NK from the reference: the CD8 estimate jumps from 12% to 22% (truth 13%) — NK's share is dumped on the kin that shares the cytotoxic module, and without the truth it looks flawless. Before unmixing, ask: does the reference have a column for every major type



Trap in one line

Proportions are compositional: only CD8 expands fourfold, yet every other type "appears" to drop — always phrase in shares, CLR before cross-sample comparison; porting across platforms is a systematic bias no sample size can rescue

B18 • Ecosystem and finale: multi-omics, spatial, Python

In one line: what you take away is not nineteen episodes of commands but the discipline of looking at the data before judging



The modality map and WNN

CITE-seq, scATAC, spatial and TCR/BCR each patch one blind spot of the transcriptome; WNN weights are per cell — T subsets resolved by ADT while B cells are identified by RNA, simultaneously



A spot is mini-bulk with coordinates

A Visium spot mixes roughly 1–10 cells — B17's deconvolution equation applies directly; the routes on the map hand off to one another



Three h5ad pitfalls

Survives: X, raw, obs, obsm; drops: scale.data, the KNN graph, extra assays, factor levels. Line one after converting prints the shape — "transpose just in case" is how correct orientations get broken



Trap in one line

Treat every cross-ecosystem handoff as a moving-day inspection — the movers never tell you which box went missing: three lines checking shape, column types and layers save you three days of debugging

B19 • Strategy design for complex samples: different tumors, different playbooks

In one line: tumor is not a singular noun — ask what the sample is before deciding how to fight (suggested order: B17→B19→B18)



Five decision points

Who is the statistical unit, how far integration corrects, how malignancy is called, how the comparison is designed, which downstream road the data supports — derive the pipeline from the question, answer all five before writing code



Six scene playbooks

GBM is the solid-tumor prototype (per-patient clusters + chr7+/10- + Neftel states); melanoma's protagonist is the T cell; breast cancer's subtype comes first; liver cancer shifts the whole mt baseline; AML flips everything; multi-region/longitudinal is won at design time



AML: the full reversal

Malignant cells look normal and CNV is often clean — the one-island-per-patient signal vanishes and inferCNV fails; switch to differentiation block + mutations + immunophenotype, with a normal bone-marrow reference atlas as a necessity. The only rule that never flips: the patient is the statistical unit



Trap in one line

Copying parameters across tumors — GBM's mt < 20% ported to PBMC lets the garbage in; presets may be overturned, but overturning demands a written reason, exactly like a QC threshold

⑤ + The complex-sample strategy room: one extra stop in every episode

Every episode B1-B18 now ends its trap section with a strategy room: "The standard walkthrough is done — now imagine your sample is not PBMC but a tumor." The five decision points map to episodes below; B19 assembles them

THE ROOM



Strategy = decisions

Walkthroughs stay intact; only the decisions change

POINT 1



Who is the unit

The unit is the patient, never the cell (B12, B19)

POINT 2



How far integration goes

Align the TME; malignant cells per patient (B11, B19)

POINT 3



How malignancy is called

CNV × clusters × markers; AML switches evidence (B16, B19)

POINT 4



How to compare

Pair and batch before sampling (B4, B12)

POINT 5



Which road

Audit the cell-count table first (B13-B17, B19)

Number card 1: measured numbers from upstream and QC

Three places where the numbers move a lot: materials, interpretation, filtering (B2, B3, B5)



2,300

genes apart from the same
FASTQ under two references
(20,300 vs 22,600)



+43% → +19%

extra UMIs per doubling of reads at
32% saturation — cost grows linearly,
information logarithmically



2,700 → 2,638

PBMC 3k cells before and after
filtering: 62 cells (~2.3%) removed,
every step logged

Number card 2: three cautionary numbers from downstream

Cluster counts are asked for, UMAP distances are scenery, cell-level p values are hallucination (B8, B9, B12)



6 → 16

clusters from the same data across
resolution 0.2–1.4 — settled by the
clustree stable band plus markers, not by
feel



$\rho = -0.04$

rank correlation of all 15 inter-cluster
UMAP distances across two seeds — as
informative as a coin flip



303 vs 106

significant genes on the same simulated
data: cell-level vs pseudobulk (on real
ifnb, 3,187 vs about 1,930)

The whole pipeline at a glance: the function chain

Nineteen episodes of code on one card — every line has a full episode behind it

```
## Upstream (B2-B4): FASTQ + reference -> count matrix
cellranger mkref /m kfastq          # B2 the materials
cellranger count (multi for samples) # B3-B4 release only after web_summary

## Downstream main line (B5-B10, PBMC 3k)
Read10X() |> CreateSeuratObject()    # B5 13,714 genes x 2,700 cells
PercentageFeatureSet(); subset()    # B5 cutoffs from distributions -> 2,638
DoubletFinder                       # B5 mark, do not remove
NormalizeData(); FindVariableFeatures(); ScaleData() # B6 or SCTransform()
RunPCA()                            # B7 nPC settled at 1:10
FindNeighbors(); FindClusters()     # B8 scan 0.2-1.4, verdict 0.6 = 10 clusters
RunUMAP(fixed.seed.use)             # B9 pictures only, never statistics
FindAllMarkers(); SingleR; Azimuth  # B10 three-way cross-validation

## Multi-sample branch (B11-B12, ifnb)
split() -> IntegrateLayers() -> JoinLayers()    # B11 geometry uses integration
AggregateExpression() -> DESeq2 -> lfcShrink()  # B12 statistics returns to RNA

## Advanced downstream (B13-B19): the "so what" after the list
-> fgsea / CellChat / slingshot / infercnv / nnls # one episode each (B13-B17)
strategy_checklist(); strategy_presets          # B19 five decisions, six scenes; B18 opens the ecosystem
```

Each episode ends with saveRDS and the next opens with readRDS — the full relay tables are on the next two cards

The RDS relay (1): main line B5–B10

Every output/*.rds is a save point you can rewind to — B5's lesson: skip the raw save and rewinding costs a Cell Ranger rerun

File	Written by	Read by
pbmc_raw.rds / pbmc_b05_qc.rds	B5, pre-filter backup / QC done	B6, B7, B8
pbmc_b06_lognorm.rds (+ b06_sct.rds)	B6, normalization done (SCT for comparison)	B7
pbmc_b07_pca.rds	B7, PCA with nPC settled	B8
pbmc_b08_clustered.rds	B8, verdict 0.6, ten clusters	B9, B10
pbmc_b09_umap.rds / pbmc_b10_annotated.rds	B9 UMAP / B10 final labels in metadata	Anything downstream + B17's signature

The RDS relay (2): the branch and B-IV

B12's DE results hand off to B13; each B-IV episode keeps its own save, and the reader is your paper

File	Written by	Read by
ifnb_b11_integrated.rds	B11, integrated and canary-approved	B12, B14
ifnb_b12_de_results.rds	B12, pseudobulk DE results plus audit	B13
ifnb_b13_gsea.rds	B13, fgsea / ORA / module scores	Your paper
ifnb_b14_cellchat.rds	B14, merged CTRL+STIM CellChat object	Your paper
sim_b15_trajectory.rds	B15, simulated differentiation + pseudotime	Your paper
b16_infercnv_obj.rds	B16, inferCNV object + malignancy labels	Your paper
b17_deconv_results.rds	B17, deconvolution estimates + ground truth	Your paper

The most expensive trap in the series: treating cells as biological replicates

What it is

Running FindMarkers on stim vs ctrl treats 30,000 cells as 30,000 independent observations — when the experiment has 16 people's samples.



Why it happens

Test power explodes with n , but cells from one donor are correlated; donor-to-donor variation gets misread as a condition effect, and even null genes collect astronomical p values.

What it costs you

P values hit $1e-320$ and the significant list inflates by tens of percent to an order of magnitude (303 vs 106; 3,187 vs ~1,930). The fix is one line: AggregateExpression sums cells back into samples, then DESeq2 takes over.

Iron rules of the whole series (1)

Every rule here was paid for in at least two episodes

RULE 1

Distributions first, thresholds second

Your data's property, not a constant. Record the reasons. (B5)



RULE 2

Seed nailed, versions locked

`set.seed(1234) + renv + relative paths = reproducible.` (B1, B9)



RULE 3

QC is a loop, not a gate

Debris appears after clustering; go back and redraw. (B5, B8)



RULE 4

Analysis in PC, plots on UMAP

Clustering, distances, DE: never on UMAP coordinates. (B8, B9)



RULE 5

Geometry integrates, stats on RNA

Integration is borrowed alignment; DE on RNA counts. (B11, B12)



RULE 6

Cells are not replicates

The unit is the sample; pseudobulk keeps n honest. (B12, A1)



Iron rules of the whole series (2)

RULE 7

When in doubt, take more PCs



The cost of too many is a little noise and a few seconds; the cost of too few can be a cell subset you never learn existed. (B7)

RULE 8

Loud errors are a gift



Silent errors (a missing GTF entry, unnormalized depth, debris clusters) are caught only by checks turned into rituals — grep, cor, the per-cluster QC table. (B2–B8)

RULE 9

Clustering hands you hypotheses



Cluster counts are made to order, labels reshuffle, junk has marker lists too; only stability + markers + QC + doublets together promote a cluster to a finding. (B8)

RULE 10

A score is not a name



P values, similarities and prediction scores are evidence, not answers; marker resolution must match cluster resolution, and the signature on the label is always yours. (B10)

Iron rules of the whole series (3): three added by B-IV and the strategy rooms

The shared discipline of the advanced episodes and the strategy rooms: braking well is worth more than accelerating well

RULE 11



Inference, not observation

Communication and trajectories: the tool always answers; only conclusions that survive a second granularity or root count, and the verb is always "suggests". (B14, B15)

RULE 12



The reference decides everything

inferCNV's baseline, deconvolution's signature: the wrong choice throws no error. Ask what the reference stands on before running; inspect its rows first after. (B16, B17)

RULE 13



A parameter is a hypothesis: a number, a reason, a record

Defaults answer someone else's data; copying across tumors is just as risky. Every threshold gets a number and a reason — so does overturning one. (B5, B19, the rooms)

TRACK B • IN ONE SENTENCE

**From FASTQ to a gene list, and on to the "so what"
beyond it —
nineteen episodes, and you walked the whole road
yourself.**

Commands go stale; the judgment — distributions first, thresholds second, reasons written down — does not. Tools get replaced, you remain.

WHAT COMES NEXT

Series C • The maths behind it (8 episodes)

Three roads onward: maths, engineering, biology

The maths road → Series C: PCA's SVD, clustering's modularity, UMAP's objective, computed step by step.

The engineering road → pipelines (targets/Nextflow) and reproducible Quarto reports; the biology road → the atlas papers of your tissue and disease, plus cellxgene census. Wherever you last got stuck, patch that first.